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Fuzzy Aware Block Fusion Deeplabv3+ for High-Resolution Remote Sensing Images Segmentation

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***Abstract*—High-resolution remote sensing images (HRRSI) provide detailed spatial information, but complex scene details often blur class boundaries and increase segmentation ambiguity. In addition, the complex spatial environment and equipment factors can lead to various noises in the images, which affect the segmentation results. To address these issues, we propose FA-DeepLabV3+, which integrates a fuzzy-aware structure with DeepLabV3+. The fuzzy aware structure (FAS) is designed, which employs a gaussian blur function to adaptively identify segmentation boundaries, effectively mitigating boundary ambiguity. Aiming at the problem of weak target and background noise interference, the attention mechanism is introduced to enhance the expression ability of key target features. A lightweight MobileNetV2 network is used to balance the accuracy and memory consumption problems. Meanwhile, the model introduces multiple skip connections to integrate shallow features, further refining the delineation of object boundaries. Experiments conducted on the Gaofen image dataset (GID) demonstrate that FA-DeepLabV3+ outperforms existing mainstream segmentation methods, achieving mIoU of 76.20% and OA of 87.77%. Compared to the DeepLabV3+ model, the proposed approach delivers superior segmentation performance and significantly reduces computational cost.**

Keywords—remote sensing images, DeepLabv3+ model, fuzzy logic, semantic segmentation, attention module

I. INTRODUCTION

Recent advancements in satellite technology have greatly enhanced the acquisition of high-resolution remote sensing images (HRRSI). Accurate semantic segmentation of HRRSI is essential for intelligent interpretation of remote sensing data [1]. However, HRRSI poses challenges like dense objects, large-scale variations, complex backgrounds, and class imbalance, increasing segmentation difficulty. Currently, deep learning neural networks are the predominant approach. U-Net [2] enhances spatial detail by introducing skip connections between the encoder and decoder layers. Furthermore, techniques such as multi-scale feature fusion [3] and attention mechanisms [4] have further improved segmentation performance. In particular, the DeepLabV3+ model leverages the atrous spatial pyramid pooling (ASPP) module to capture image features at multiple scales, improving robustness in object recognition and segmentation. However, challenges such as inter-class confusion and blurred boundaries in HRRSI introduce significant uncertainty in segmentation results.

Fuzzy set theory reduces uncertainty in feature representation through fuzzy modeling. Deep neural networks excel at extracting features from raw data. Recently, deep learning networks based on type-2 fuzzy logic systems have gained attention in remote sensing applications. Zhou et al. [5] employed neural networks to extract low-dimensional features and utilized fuzzy features for classification. However, these multi-stage methods are difficult to train end-to-end. To address this, some researchers have combined fuzzy logic with a convolutional neural network (CNN) to enable end-to-end trainable segmentation models. Wei et al. [6] proposed a multi-branch fuzzy convolutional neural network architecture to address uncertainty at class boundaries. Nan et al. [7] improved fuzzy logic networks for airway segmentation by using an "OR" aggregation operator to enhance performance.

This paper proposes a DeepLabV3+ model with fused fuzzy structure (FA-DeepLabV3+). It incorporates a MobileNetV2 backbone, attention mechanisms, and multi-level skip connections to enhance segmentation accuracy and efficiency. The contributions are summarized as follows:

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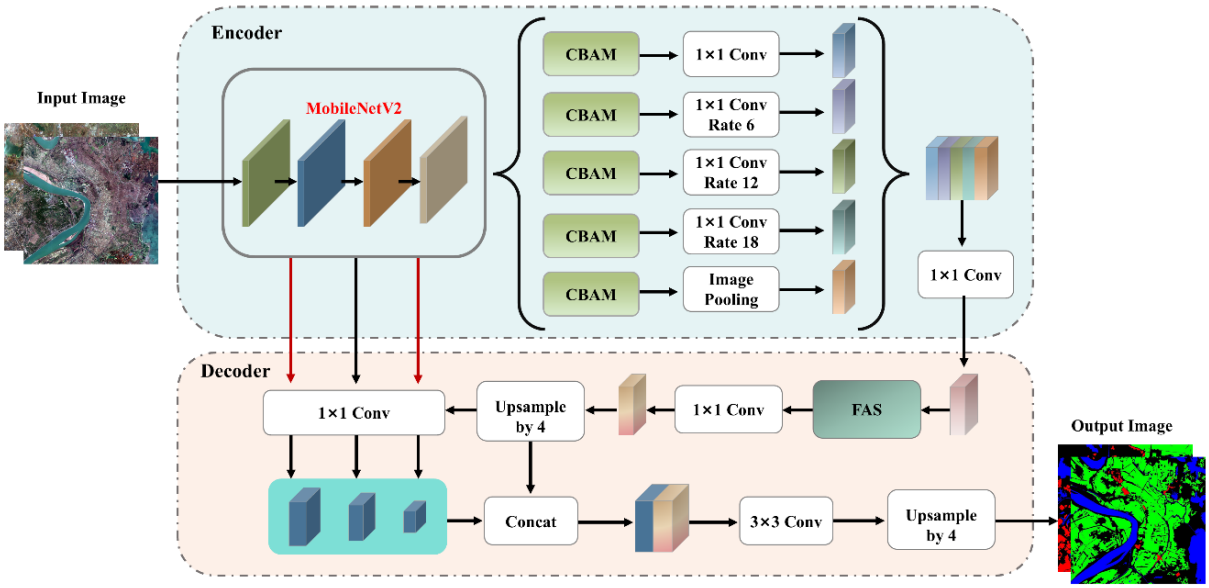


Fig. 1. Structure of the FA-DeepLabv3+ network.

(1) A fuzzy aware structure (FAS) is designed to adaptively reduce boundary uncertainty in remote sensing segmentation, thereby enhancing accuracy and robustness.

(2) We propose an FA-DeepLabV3+ network based on multi-scale context information, adopting MobileNetV2 as the backbone for efficient semantic segmentation of HRRSI. Segmentation performance is further improved through adaptive feature fusion via multiple skip connections in the DeepLabV3+ architecture.

(3) Experimental results on the Gaofen image dataset (GID) show that the FA-DeepLabV3+ network achieves state-of-the-art segmentation performance. The model has the advantages of being lightweight and low computational cost, and performs well in segmentation accuracy and efficiency.

II. Methodology

A. Overview

The architecture of the FA-DeepLabV3+ network is shown in Fig. 1. It inherits the encoder-decoder design of the original DeepLabV3+, with MobileNetV2 as the encoder backbone for efficient extraction of deep semantic features. This ensures segmentation accuracy while reducing model parameters and accelerating training convergence. Before the ASPP module, the convolutional block attention module (CBAM) is integrated to fully exploit the key information of channel and spatial dimensions. To address the uncertainty and complexity of feature boundaries, this paper combines deep learning methods with fuzzy logic to design FAS. This module aims to model the intrinsic connection between feature maps and segmentation results, thus reducing data uncertainty and improving the segmentation performance for small-scale targets and boundary regions. The decoder includes upsampling and convolution operations, with multiple skip connections between the encoder and decoder to efficiently fuse deep and shallow features, improving segmentation clarity and accuracy. Finally, the feature map is reduced to its original resolution by the upsampling operation to output the final segmentation prediction results.

B. Convolutional Block Attention Module

FA-DeepLabV3+ uses CBAM to improve segmentation of tiny targets and complex boundaries in HRRSI. CBAM enhances key features and suppresses irrelevant ones. It has two parts: channel attention module (CAM) and spatial attention module (SAM). CAM weights channels relevant to segmentation. SAM highlights important spatial regions. They work in sequence to boost feature representation.

C. Fuzzy Aware Structure

As shown in Fig. 2, the input feature map of the FAS module is noted as $F \in R^{C\times H\times W}$, where H and W represent the height and width of the feature map, respectively, and C is the number of channels. Given that the Gaussian membership function (GMF) has good smoothness and concise expression ability, we propose a trainable GMF for constructing fuzzy sets. Each element in the fuzzy set determines its degree of belonging in the set based on its affiliation degree. The initialization of GMF parameters uses gaussian distribution random initialization, with a mean of 0 and a standard deviation of a small positive number. This is used to introduce a certain degree of randomness in the early stages of network training, promoting parameter diversity and thereby improving the convergence and stability of the network. For each feature point of a particular channel $\alpha \in R^{C\times W\times H}$, it is filtered by m GMF. The specific expression of GMF is shown in (1):

$$f_{i,j}(X,\mu,\sigma) = e^{\frac{-(X_j-\mu_{i,j})^2}{2\sigma_{i,j}^2}} \tag{1}$$

where $i = 1,\dots,m, j = 1,\dots,C, X_j$ denote the coordinates of the feature points in the channel, $u_{i,j}$ and $\sigma_{i,j}$ are denoted as the mean and standard deviation of the ith GMF in channel j, respectively.

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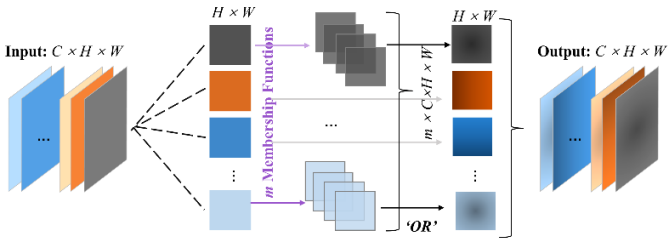


Fig. 2. Structure of the FAS.

Unlike traditional approaches that typically employ the "AND" operator, we utilize GMF to adaptively learn the importance of each feature representation. Specifically, the "OR" operator is introduced and applied to all affiliation values of each feature point α. This approach effectively suppresses irrelevant or weak features while highlighting those with significant contribution to the segmentation task. Assuming that there are two fuzzy sets, $\bar{A}$ and $\bar{B}$, the mathematical expression using the "OR" operation is shown as (2):

$$f_{\bar{A}\cup\bar{B}}(y) = f_{\bar{A}}(y) \vee f_{\bar{B}}(y), \forall y \epsilon U \tag{2}$$

where U denotes the full set and y is an element of U. The "OR" operation is discrete and non-differentiable. However, the max operation is continuous and differentiable, enabling end-to-end backpropagation training. Thus, the formula is revised to (3):

$$f_{\bar{A}\cup\bar{B}}(y) = max\,(f_{\bar{A}}(y), f_{\bar{B}}(y)) \tag{3}$$

Therefore, according to (2) and (3), the affiliation degree of the jth channel can be obtained $f_j(X, \mu, \sigma)$, where $\vee$ denotes the joint operation, as shown in (4):

$$f_j(X,\mu,\sigma) = \bigvee_{i=1}^{m} e^{\frac{-(X_j-\mu_{i,j})^2}{2\sigma_{i,j}^2}} = max\,(e^{\frac{-(X_j-\mu_{i,j})^2}{2\sigma_{i,j}^2}}) \tag{4}$$

III. Experimental Results

A. Experimental Settings

The GID is a large-scale, high-resolution land cover remote sensing dataset. It is divided into five foreground categories: build-up, farmland, forest, meadow, and water, and one background category "other". In this study, 28 images were selected for training and 11 for testing. All experiments were implemented using PyTorch on NVIDIA RTX 4090D GPUs. Model performance was evaluated using mIoU, F1-Score, and OA metrics.

B. Analysis of Results

For comprehensive evaluation, we conducted comparative experiments between the proposed FA-DeepLabV3+ and several mainstream segmentation methods on the GID dataset, including UNet [2], MANet [8], BANet [9], CMTFNet [10], and DeepLabV3+ [3] models. TABLE I. demonstrates the quantitative results. The experimental results showed that FA-DeepLabV3+ achieved optimal performance on all evaluation metrics, specifically obtaining 76.20% of mIoU, 64.62% of F1-Score, and 87.77% of OA. Compared with the DeepLabV3+ model, FA-DeepLabV3+ improved 2.12%, 2.81% and 4.11% on F1-Score, OA, and mIoU, respectively. Computational complexity was evaluated using floating point operations per second (FLOPs), parameter count, and weight size. DeepLabV3+ had 43.44G FLOPs, while FA-DeepLabV3+ had only 28.39G, a reduction of 15.05G. Additionally, DeepLabV3+ had a parameter file size of 226 MB and a weight size of 59.34 MB, whereas FA-DeepLabV3+ required only 23.3 MB and 6.06 MB, respectively. These reductions highlighted the lightweight nature of the FA-DeepLabV3+ model.

To evaluate model performance in HRRSI segmentation, representative GID samples were selected for analysis. As shown in Fig. 3, FA-DeepLabV3+ exhibited superior contour prediction in farmland and meadow regions due to multi-scale feature fusion. It excelled in spatial connectivity and edge integrity in the third scene. Other models frequently misclassified regions as farmland or forest in the second, fourth, and sixth scenes, whereas FA-DeepLabV3+ delivered more accurate and smoother boundaries by capturing local semantic details. Additionally, it outperformed competitors in segmenting farmland and built-up areas in the fifth scene. Overall, the proposed model showed significant advantages in segmentation accuracy and boundary refinement in complex scenes.

Ablation results are presented in TABLE II. DeepLabV3+ served as the baseline model, with functional modules added progressively. CBAM and FAS complemented each other by enhancing spatial-channel features and modeling uncertainty in remote sensing data, respectively. Their combined use better demonstrated performance improvements in practical scenarios. As this study focused on overall segmentation performance, detailed analysis of module interactions was omitted. Experiments showed that adding multiple skip connections improved F1-Score, OA, and mIoU by 0.35%, 0.75%, and 2.59%. After introducing the CBAM attention module, F1-Score, OA, and mIoU were improved by 1.02%, 1.86% and 2.99%, respectively, over the baseline model. After further integrating the FAS structure, F1-Score, OA, and mIoU were improved by 2.12%, 2.81% and 4.11% respectively, compared to the baseline model. These results confirmed that the modules effectively enhanced the semantic segmentation accuracy of HRRSI.

TABLE I. Quantitative Comparison Results on GID. The best values are in bold.

Method	Build-up	Farmland	Forest	Meadow	Water	F1-Score(%)	OA(%)	mIoU(%)
UNet	65.00	81.72	69.02	50.55	91.08	61.68	85.12	70.81
MANet	65.75	81.56	72.28	62.09	91.65	63.26	85.95	73.51
BANet	64.63	**83.49**	68.99	63.79	89.24	63.03	86.02	73.00
CMTFNet	62.97	77.38	**79.03**	62.99	90.90	63.10	84.76	73.20
DeepLabV3+	**66.08**	78.32	70.70	57.84	**92.32**	62.50	84.96	72.09
FA-DeepLabv3+	65.66	83.43	78.65	**65.27**	91.94	**64.62**	**87.77**	**76.20**

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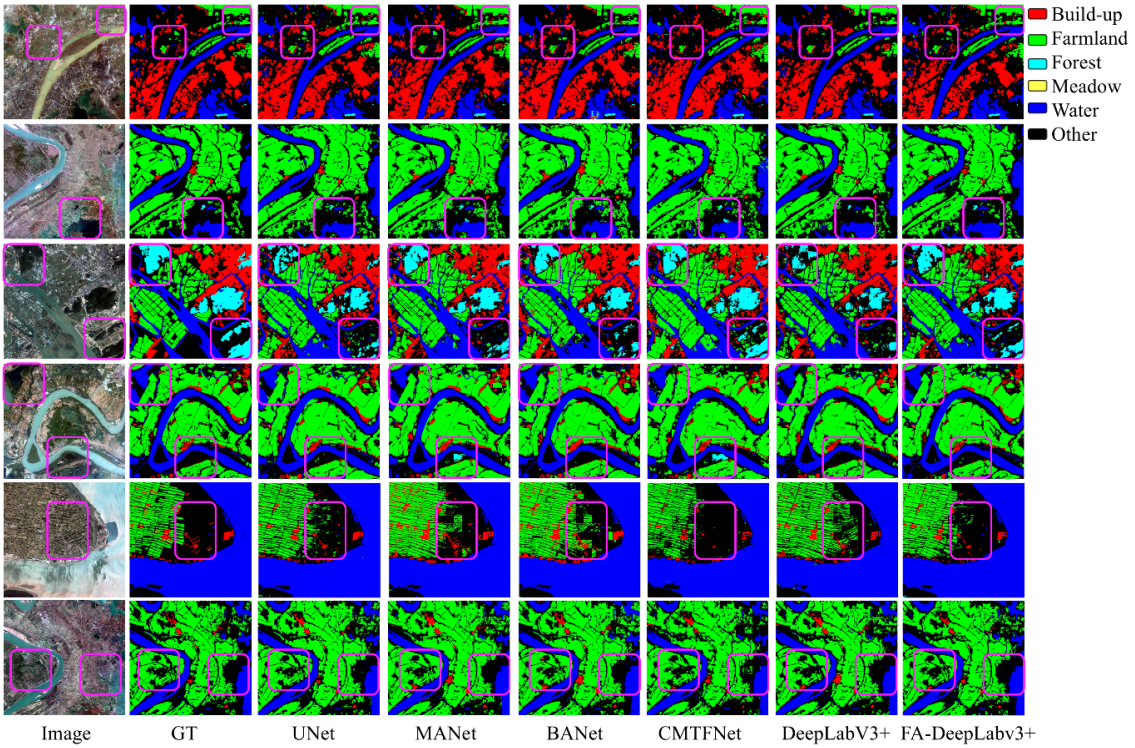


Fig. 3. Segmentation visualization results with different methods on the GID. The pink boxes indicate where our method clearly outperforms the others.

TABLE II. ABLATION STUDY RESULTS ON GID. THE BEST VALUES ARE IN BOLD.

Multiple Skip Connections	CBAM	FAS	F1-Score(%)	OA(%)	mIoU (%)
Baseline Network			62.50	84.96	72.09
√			62.85	85.71	74.68
√	√		63.52	86.82	75.08
√	√	√	**64.62**	**87.77**	**76.20**

IV. CONCLUSION

This paper proposes FA-DeepLabV3+, an improved DeepLabV3+ with fuzzy structure and lightweight modules for HRRSI segmentation. It integrates attention and FAS to enhance key feature extraction and reduce boundary uncertainty. Multiple skip connections enable efficient feature fusion and finer boundaries. Experiments on GID show improved accuracy with reduced complexity. Ablation studies confirm CBAM and FAS effectiveness, especially for small targets. Future work will optimize the network, extend to multi-source data, and pursue more efficient and generalizable models.

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REFERENCES

[1] Y. H. Liu, Y. F. Zhang, Y. Wang, and S. H. Mei, "Rethinking transformers for semantic segmentation of remote sensing images," IEEE Trans. Geosci. Remote Sens., vol. 61, pp. 1–15, 2023.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in Medical Image Computing and Computer-Assisted Intervention, Munich: Germany, 2015, pp. 234–241.

[3] L. C. Chen, Y. K. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in Proceedings of the European Conference on Computer Vision, Munich: Germany, 2018, pp. 801–818.

[4] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "Cbam: Convolutional block attention module," in Proceedings of the European Conference on Computer Vision, Munich: Germany, 2018, pp. 3–19.

[5] S. S. Zhou, Q. Q. Chen, and X. L. Wang, "Fuzzy deep belief networks for semi-supervised sentiment classification," Neurocomputing, vol. 131, pp. 312–322, 2014.

[6] G. Y. Wei, J. D. Xu, Q. P. Chong, J. J. Huang, and H. H. Xing, "Prior-guided fuzzy-aware multibranch network for remote sensing image segmentation," IEEE Geosci. Remote Sens. Lett., vol. 9, no. 4, 2024.

[7] Y. Nan, J. Del Ser, Z. Y. Tang, P. Tang, X. D. Xing, Y. Y. Fang et al., "Fuzzy attention neural network to tackle discontinuity in airway segmentation," IEEE Trans. Neural Netw. Learn. Syst., vol. 35, no. 6, pp. 7391–7404, 2023.

[8] P. He, L. C. Jiao, R. H. Shang, S. Wang, X. Liu, D. Quan et al., "MANet: Multi-scale aware-relation network for semantic segmentation in aerial scenes," IEEE Trans. Geosci. Remote Sens., vol. 60, pp. 1–15, 2022.

[9] L. B. Wang, R. Li, D. Z. Wang, C. X. Duan, T. Wang, and X. L. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," Remote Sens., vol. 13, no. 16, pp. 3065, 2021.

[10] H. L. Wu, P. Huang, M. Zhang, W. L. Tang, and X. Y. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," IEEE Trans. Geosci. Remote Sens., vol. 61, pp. 1–12, 2023.

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